# SAKIB CHOWDHURY

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## SUMMARY

Robotics and machine learning researcher specializing in generative policies and learning from human demonstrations for robotic manipulation. Experienced in multimodal policies, world models and force-conditioned policy learning, with a focus on learning generalizable robot behaviors. Developing and deploying learning-based policies across simulation and real robotic platforms.

## EDUCATION

**Ph.D. in ECE**, *CGPA: 3.97/4.00* (2023 - Present) Stevens Institute of Technology
Research Focus: Learning multimodal behaviors from human demonstrations using generative policies.

**M.Sc. in ECE (Robotics Concentration)**, *CGPA: 3.97/4.00* (2023 - 2025) Stevens Institute of Technology
Relevant Coursework: 3D Computer Vision, Applied Modeling & Optimization; Autonomous Navigation for Mobile Robots; Mobile Manipulators; Wearable Robotics;

**B.Sc. in EEE** , *CGPA: 3.65/4.00* (2017 - 2022) Bangladesh University of Engineering and Technology (BUET)
Relevant Coursework: Linear Control Systems; Digital Logic Design; Digital Signal Processing, Microprocessors, C Programming;

## RESEARCH EXPERIENCE

**Graduate Researcher (Robotics & Automation Lab)** Stevens Institute of Technology, NJ, USA
*Sept 2023 – Present*

- Developed a **force-conditioned generative visuomotor policy for robotic tool use from human demonstrations**, introducing robot-replay-based external force capture from visual demonstrations and onboard joint torque sensing, eliminating the need for external F/T sensors. Used gaussian splatting to render novel views of the scene. Demonstrated generalizability across **diffusion, ACT, and flow-matching policies**, with force-based admittance correction improving manipulation performance. [LINK]
- Developed a **neural motion planner for time critical tasks**. showed that joint space trajectories generated by tree-based classical planners are piecewise linear in nature, and learning the breakpoints of these piecewise linear trajectories can make the planner faster, **achieving 6.5 ms average planning time** for rapid-response tasks such as robotic ball catching. [LINK]
- Developed a **residual learning method for projectile trajectory prediction** to capture unmodeled dynamics such as friction, air resistance, and restitution; demonstrated improved interception performance in a simulated robotic table-tennis environment using PyBullet. [LINK]

**ML Research Engineer** Celloscope, Dhaka, Bangladesh
*Dec 2021 – Aug 2023*

- Developed **Bangla automatic speech recognition (ASR) and text-to-speech (TTS) systems**, fine-tuning Wav2Vec and Whisper models and training a **Conformer-based ASR** using ESPnet/Kaldi-style recipes; achieved **21% WER on raw production audio, outperforming Wav2Vec and Whisper** on Bangla; Trained a **Tacotron-based TTS** system for Bangla; [LINK]
- Built an **audio-to-dialogue extraction pipeline** combining speaker diarization, ASR, and downstream processing to automatically transform multi-speaker audio into structured dialogue.
- Designed a **data collection and annotation pipeline using Label Studio** for image-to-text extraction, supporting dataset creation and model development for document and license-plate recognition.
- Trained and deployed **DONUT, a document understanding Vision Transformer (ViT)**, for Key Information Extraction (KIE) from **Bangladeshi national ID cards and vehicle license plates**; [LINK]

## SELECTED PROJECTS

- **Gaussian Splatting of a Robot Arm**: Reproduced a Gaussian-splatting-based simulation pipeline for rendering novel views of a real robotic workspace while moving the robot in simulation. Integrated real-world workspace reconstruction with simulated robot interaction to improve visual realism and reduce the sim-to-real gap for future learning-based robotic manipulation research. [LINK]
- **Synthetic Episode Generation using World Model**: Used NVIDIA Cosmos3 video world model to generate synthetic videos for tool usage policy learning for robotic manipulators. [LINK]
- **Bangla Voice Banking System**: Built a Bangla speech-to-speech system integrating ASR, TTS, and LLAMA; trained ASR and TTS models, and used LLAMA for banking specific FAQs. [LINK]
- **CovTANet**: We developed a novel triple attention based neural network architecture to diagnose and segment lesions from chest CT scans. This architecture can process 3D volumetric images without using 3D convolutions. [LINK]
- **Automated Vehicle Tracking at Toll Plazas**: Built a real-time vehicle tracking and license-plate recognition system—YOLO for plate detection, Vision Transformers (ViT) for number extraction — deployed at Kalna Toll Plaza, Bangladesh for live identification and monitoring. [LINK]

## TECHNICAL SKILLS

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|---|---|
| **ML Frameworks** | PyTorch, TensorFlow, Scikit-Learn, Numpy |
| **Robotics Softwares and Simulators** | ROS, Bullet, IsaacSim, Gazebo, OMPL, MoveIt |
| **Programming Languages** | Python, C/C++, Bash, MATLAB, Simulink |
| **Embedded Microcontrollers** | ATmega328, STM32, ESP32, MSP430 |
| **Others** | Linux, Docker, Git, Flask, OpenCV, FastAPI, PostgreSQL |

## SELECTED PUBLICATIONS

- **S. Chowdhury**, R. Huang, V. Chen, Y. Guo, "Force-Conditioned Imitation Learning from Vision-Only Human Demonstrations via Robot Replay", (Submitted at ICRA 2027). [LINK]
- **S. Chowdhury**, Y. Guo, "FlashPlan: A Fast Neural Motion Planner for Robotic Ball Catching", (Submitted at ICRA 2027). [LINK]
- **S. Chowdhury**, Y. Guo, "Learning to Strike for Robotic Table Tennis". [LINK]
- **S. Chowdhury**, M. Morshed, S. A. Fattah, "SpectroCardioNet: Attention-Based Triple-Spectrogram PCG Network for Cardiac Disease Detection", IEEE Sensors Journal, 2022. [LINK]
- **S. Chowdhury**, D. K. Sikder, A. Roy, "A Simulated Intelligent Pixelated Electrode Array for Surface Electromyography Sensors", IEEE Sensors Journal, 2023. [LINK]
- S. M. Monsur, Shariar Kabir, **S. Chowdhury**, "SynthNID: Synthetic Data for Bangla Key Information Extraction", EMNLP Bangla Workshop 2023. [LINK]
- S. M. Monsur, **S. Chowdhury**, M. S. Fatemi, et al., "SHONGLAP: A Large Bengali Open-Domain Dialogue Corpus", LREC 2022. [LINK]
- T. Mahmud, M. J. Alam, **S. Chowdhury**, et al., "CovTANet: Hybrid Tri-Level Attention Network for COVID-19 Lesion Segmentation and Severity Prediction", IEEE Transactions on Industrial Informatics, 2021. [LINK]